DEVELOPMENT OF A WEB-BASED INSTITUTIONAL APPOINTMENT AND VISITOR MANAGEMENT SYSTEM WITH SEMI-AUTOMATED SCHEDULING, DEPARTMENT-BASED APPROVAL, AND EMAIL NOTIFICATIONS AT SYSTEMS PLUS COMPUTER COLLEGE

A Capstone Proposal

Systems Plus Computer College
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CHAPTER 1
INTRODUCTION

Background of the Study

The increasing use of information and communication technology has encouraged educational institutions to improve administrative coordination and service delivery through digital platforms. In colleges, appointments occur not only with external visitors but also among students, faculty members, and institutional departments. Academic consultations, administrative inquiries, document-related concerns, and meetings require clear schedules, responsible personnel, and accurate communication.

Institutional appointment management becomes more complex when multiple departments maintain separate availability schedules and approval procedures. Requests submitted through different channels can be difficult to consolidate, and users may be uncertain about the status of their transactions. A centralized web-based platform offers a way to organize appointment requests, communicate decisions, and maintain records for authorized personnel.

Systems Plus Computer College (SPCC) in Caloocan City is the intended setting of this study. Its students, faculty members, and administrative departments represent distinct user groups with different appointment requirements. External visitors may also need to arrange visits with designated offices. The researchers will validate the current procedures, challenges, and approval responsibilities through interviews, observation, and requirements gathering; the existence or frequency of specific problems will not be presumed without institutional evidence.

The proposed system adopts semi-automated scheduling, in which users select from department-configured availability and the system assists with request organization and conflict checking. Final decisions remain with authorized departmental personnel. Email notifications will communicate request submissions, approval or rejection decisions, and relevant schedule changes. Appointment records and visitor information will be accessible according to assigned roles and institutional policies.

Because the application will process personal information, its design must consider the Data Privacy Act of 2012 (Republic Act No. 10173, 2012), appropriate access controls, and secure data handling. The system may also be evaluated using selected characteristics of the ISO/IEC 25010:2023 software product quality model (ISO/IEC, 2023). Technical guidance from

NIST (2025), OWASP (2025), and W3C (2024) can inform authentication, application security, and accessible interfaces.

Therefore, this study proposes the development of a Web-Based Institutional Appointment and Visitor Management System with Semi-Automated Scheduling, Department-Based Approval, and Email Notifications for Systems Plus Computer College. It aims to support coordinated appointment requests across participating departments while preserving human approval authority and enabling structured visitor and appointment record management.

Statement of the Problem

This study seeks to determine how a centralized web-based appointment and visitor management system can support the scheduling and coordination requirements of Systems Plus Computer College. Specifically, it aims to answer the following questions:

- How can students, faculty members, department personnel, and authorized external visitors submit and monitor institutional appointment requests through a centralized platform?
- How can semi-automated scheduling present available appointment slots and identify potential conflicts while retaining final approval by designated personnel?
- How can a department-based approval workflow organize the review, approval, rejection, and management of appointment requests?
- How can automated email notifications communicate appointment submissions, decisions, cancellations, and schedule changes?
- How can the system maintain appointment and visitor records with appropriate role-based access restrictions?
- How can the developed system be evaluated in terms of selected ISO/IEC 25010:2023 software quality characteristics?

Objectives of the Study

General Objective

To design, develop, and evaluate a web-based Institutional Appointment and Visitor Management System for Systems Plus Computer College – Caloocan that supports students, faculty members, participating departments, administrators, and authorized visitors through semi-automated scheduling, department-based approval, and email notifications.

Specific Objectives

- To design and develop a centralized portal for submitting, viewing, and tracking institutional appointment requests.
- To implement department-configured availability and semi-automated scheduling with conflict checking and human approval.
- To establish department-based review and approval functions for designated personnel.
- To integrate email notifications for appointment submission, approval, rejection, cancellation, and rescheduling.
- To develop appointment history, visitor registration, and administrative reporting functions with role-based access control.
- To evaluate the developed system using selected ISO/IEC 25010:2023 quality characteristics and feedback from representative SPCC users.

Scope and Limitations of the Study

This study covers the design, development, and evaluation of a web-based appointment and visitor management application intended for Systems Plus Computer College – Caloocan. The intended users include students, faculty members, designated department personnel, administrators, and authorized external visitors. The specific departments included in implementation will depend on institutional approval and requirements validation.

The major system components and functionalities include:

- User Portal – Allows authorized users to submit appointment requests, view their request status, and access appointment history.
- Department Scheduling – Displays availability defined by participating offices and assists in preventing overlapping approved bookings.
- Approval Workflow – Allows designated departmental approvers to review, approve, reject, or reschedule assigned requests.
- Email Notification – Sends status updates and schedule-related messages through a configured email service.
- Visitor Management – Maintains authorized external visitor appointment details and, when adopted by the institution, entry and exit records.

- Administration and Security – Supports account permissions, department configuration, reports, and access restrictions.

Limitations of the Study

- Institutional Boundary: The study is limited to SPCC – Caloocan and participating departments; it does not automatically extend to other campuses.
- Human Approval: The system does not independently approve appointment requests; designated personnel retain decision-making authority.
- Platform: Access requires a compatible web browser and network connection.
- Email Dependency: Notification delivery depends on email-service availability and accurate recipient addresses.
- Physical Security: Biometric identification, facial recognition, and automated door-access systems are excluded.
- Operational Evaluation: The study evaluates defined system functions and selected quality characteristics; long-term institutional outcomes require separate evidence.

Significance of the Study

SPCC Administration. The system may provide consolidated appointment monitoring and reporting to support institutional coordination.

Department Heads and Personnel. Department-specific schedules and approvals may help organize incoming requests and reduce repetitive follow-up work.

Students. A single portal may make it easier to request consultations or administrative appointments and check their status.

Faculty Members. The system may support the coordination of consultations and meetings according to approved schedules and permissions.

External Visitors. The application may provide a structured process for requesting visits and receiving decisions before arriving on campus.

Future Researchers. The study may serve as a reference for developing and evaluating institutional appointment and visitor management systems.

CHAPTER 2

CONCEPTUAL FRAMEWORK

Review of Related Literature and Studies

Web-Based Institutional Appointment Management

Web-based appointment management systems provide a shared environment for submitting, organizing, and monitoring meeting requests. In a college setting, appointment transactions may involve students requesting consultations, faculty members arranging meetings, and offices coordinating administrative services. A centralized application can standardize request information, show the current status of transactions, and make appointment histories accessible to authorized users. These capabilities represent intended system functions rather than confirmed outcomes at Systems Plus Computer College (SPCC).

For the proposed system, appointment requests will be associated with a specific department or authorized recipient. The system will record the purpose, requested date and time, and processing status. The precise departments, consultation types, and booking rules must be determined through interviews and requirements validation at SPCC Caloocan.

Semi-Automated Scheduling and Availability Management

Semi-automated scheduling combines computer-assisted appointment coordination with human decision-making. An application can display configured available time slots, check for potential overlapping bookings, and update request statuses. However, approval or rejection is performed by a designated institutional representative rather than being decided automatically.

This arrangement is appropriate when departments need to retain control over personnel availability, service hours, and appointment priorities. The proposed scheduling module will therefore assist with time-slot selection and conflict checking while allowing authorized approvers to make the final decision. Rules for pending slots, cancellations, and rescheduling must be explicitly defined to avoid inconsistent bookings.

Department-Based Approval and Role-Based Access Control

An approval workflow identifies the personnel responsible for acting on a request and records the outcome of that decision. In a multi-department institution, routing appointment requests to the relevant office can help establish accountability and prevent unrelated personnel from processing another department's transactions.

Role-based access control (RBAC) supports this design by granting system permissions according to assigned roles and responsibilities. The proposed system distinguishes student and faculty requesters, authorized department approvers, administrators, and external visitors. The NIST Digital Identity Guidelines (2025) and OWASP Application Security Verification Standard (2025) provide technical guidance relevant to authentication, secure sessions, and application access controls. Actual permission assignments will require institutional approval.

Email Notifications and Appointment Communication

Email notifications are an important communication component of digital appointment workflows. Messages can acknowledge receipt of a request and communicate approval, rejection, cancellation, or rescheduling. These notifications should identify the appointment status and essential scheduling details without unnecessarily exposing sensitive personal information.

In the proposed application, notification events will be linked to changes in appointment status. Successful delivery depends on the configured email service, valid recipient addresses, and network availability. The system should not treat sending an email as equivalent to a user having read or acknowledged it.

Visitor Registration, Privacy, and Recordkeeping

Visitor management complements institutional scheduling by recording information about external individuals who have legitimate appointments with school offices. Depending on approved procedures, the application may capture the visitor's name, purpose, destination department, appointment reference, and arrival or departure details. Visitor logging must remain distinct from physical entry authorization, which is controlled by institutional security personnel.

The Data Privacy Act of 2012 (Republic Act No. 10173) provides the legal context for processing personal information in the Philippines. Data collection should be limited to legitimate purposes, with access restrictions and appropriate retention procedures. The proposed

system will therefore incorporate permission-based record access and safeguards for appointment and visitor information.

Software Quality, Usability, and Accessibility

Quality evaluation helps determine whether the proposed application meets its documented requirements. ISO/IEC 25010:2023 provides a software product quality model that may guide evaluation of functional suitability, performance efficiency, interaction capability, reliability, and security. Selected characteristics should be matched with specific tests and evaluation instruments instead of being assumed from the presence of features alone.

Web Content Accessibility Guidelines (WCAG) 2.2 (World Wide Web Consortium, 2024) also provides a useful basis for accessible web interactions, including keyboard access, understandable labels, and readable interface elements. For SPCC users accessing the platform from different devices, responsive and accessible interfaces are relevant design considerations.

Synthesis of the Reviewed Literature and Technical Sources

The reviewed concepts and technical sources support the proposed combination of centralized appointment requests, semi-automated scheduling, department-based approval, notification services, and secure visitor recordkeeping. The distinguishing scope of this project is its intended coverage of multiple SPCC Caloocan departments and user categories rather than a single office or visitor-only process. These sources provide design and evaluation guidance; they do not independently establish that SPCC currently experiences specific operational problems or prove that the proposed application will improve efficiency. Such findings require institutional requirements gathering and system evaluation.

CHAPTER 2

CONCEPTUAL FRAMEWORK

Review of Related Literature and Studies

Web-Based Institutional Appointment Management Systems

Web-based appointment management systems help educational institutions organize scheduling and administrative transactions. These systems allow students, faculty members, and department personnel to request appointments, check available schedules, and monitor appointment statuses. Centralized platforms can reduce manual processes, prevent scheduling conflicts, and improve institutional efficiency.

Semi-Automated Scheduling and Approval Systems

Semi-automated scheduling systems combine digital scheduling with manual approval by authorized personnel. These systems allow users to select available schedules while department representatives review and approve appointment requests. This approach supports better coordination, reduces overlapping appointments, and maintains administrative control across institutional departments.

Automated Email Notifications and Visitor Management Systems

Automated email notifications improve communication by informing users about appointment confirmations, approvals, rejections, and schedule changes. Meanwhile, visitor management systems organize visitor information, appointment records, and visit purposes. Integrating these features into a centralized platform can improve recordkeeping, communication, and security within educational institutions.

Conceptual Model of the Study

The conceptual model adopts the Input–Process–Output (IPO) framework to organize the resources and requirements, development activities, and expected deliverables of the study.

INPUT	PROCESS	OUTPUT
• Institutional policies and department workflows • User roles and appointment rules • Visitor information requirements • Hardware and software resources • Privacy and security requirements	• Requirements gathering • System and database design • Scheduling and approval development • Email and visitor module integration • Testing and evaluation	• Web-based institutional appointment and visitor management system • Semi-automated scheduling • Department-based approval • Email notifications • Appointment and visitor records

Figure 2.1. Input–Process–Output Conceptual Model of the Study

INPUT

The input comprises appointment and visitor requirements gathered from participating SPCC departments, user categories and authorization rules, available development resources, and applicable institutional privacy and security requirements.

PROCESS

The process consists of analyzing institutional workflows; designing the user interfaces, database, and approval routing; developing the scheduling, notification, and visitor modules; integrating components; and evaluating the completed application.

OUTPUT

The expected output is a web-based institutional appointment and visitor management system that supports student, faculty, departmental, and authorized external visitor transactions. Actual quality and suitability will be determined through testing and stakeholder evaluation.

Operational Definition of Terms

Appointment Request – A submitted request for a meeting or institutional transaction addressed to a participating SPCC department or authorized personnel.

Department Approver – An authorized employee assigned to review and decide on appointment requests for a particular department.

Email Notification – A system-generated email informing a requester or responsible personnel about an appointment event or status change.

Institutional Appointment – A scheduled academic or administrative meeting or transaction involving an SPCC office or authorized representative.

Role-Based Access Control (RBAC) – A mechanism that restricts system functions and information according to assigned user permissions.

Semi-Automated Scheduling – Computer-assisted schedule selection and conflict checking with final approval performed by authorized personnel.

Visitor Record – Information maintained about an authorized external visitor and the associated institutional appointment.

Web-Based System – An application accessed using a compatible web browser over a network.

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